THE AFFINE BLOWUP ALGEBRA OF A DOMAIN IN AN IDEAL AT A NONZERO ELEMENT IS AGAIN A DOMAIN

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Description of elements of the affine blowup algebra R[I/a]. Used at the start of the counted proof (lines 17546-17547). Lets an arbitrary element be written as x/a^n with x in I^n, which is the form the argument multiplies in.
- The relation a^N x y = 0 in the blowup algebra for large N. Used at line 17548. Converts vanishing of a product in R[I/a] into an honest product relation in R, where domain-ness can be applied.

Lemma 0.1. *Let R be a domain, $I \subset R$ an ideal, and $a \in I$ a nonzero element. Then the affine blowup algebra $R[\frac{I}{a}]$ is a domain.*

Proof. Suppose x/a^n, y/a^m with $x \in I^n$, $y \in I^m$ are elements of $R[\frac{I}{a}]$ whose product is zero. Then $a^N xy = 0$ in R. Since R is a domain we conclude that either $x = 0$ or $y = 0$. □

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